

**Contact:** Science Park 904  
1098 XH Amsterdam, NL

m.klinger@uva.nl  
<https://maklinger.github.io/>

## Education

01/2020 to 11/2024 **PhD in Theoretical Astroparticle Physics with distinction (summa cum laude)**  
DESY Zeuthen  
Within the [Helmholtz Weizmann Research School on Multimessenger Astronomy](#)  
Supervisors: Walter Winter, Andrew Taylor, Eli Waxman  
Focus: *Time-Dependent Radiation Modelling of Gamma-Ray Burst Afterglows*

- Reduced analytical afterglow modelling
- Time-dependent afterglow modelling of gamma-ray bursts
- Development, maintenance and publication of the [AM<sup>3</sup> code](#)
- Counts-level fits from keV-TeV energies

10/2016 to 12/2019 **Master of Science in Physics with distinction**  
Focus: Astroparticle Physics and Cosmology  
RWTH Aachen University  
Supervisors: Thomas Bretz, Philipp Mertsch  
Thesis: *Developing a Magnetic Reconnection Model to Interpret Blazar Variability as Measured with the FACT Telescope*

10/2013 to 10/2016 **Bachelor of Science in Physics with distinction**  
RWTH Aachen University  
Supervisor: Christopher Wiebusch  
Thesis: *Development of a measurement setup to compare acoustic emitters for the EnEx-RANGE project*

## Professional Experience

10/2024 to present **Postdoctoral Researcher**  
Anton Pannekoek Institute for Astronomy / University of Amsterdam  
Advisors: Sera Markoff  
Focus: *Improving multi-messenger physics in GRMHD simulations of black holes*

06/2022 **Summer School - Foundations of Cosmic Ray Astrophysics, Varenna (Italy)**

05/2017 - 02/2019 **Assistant Scientist at EnEx RANGE Project at RWTH Aachen**  
Focus: Experimental lab work and data analysis  
Glacier expedition for project validation (08/2017)

02/2017 - 03/2017 **Site visit of Pierre-Auger-Observatory, Argentina**  
Participation in maintenance, shifts, collaboration meeting and public outreach work at the local visitor's facility

09/2016 - 12/2016 **Scientific stay at the HAWC Observatory, Mexico**  
Studies on Radio Galaxies at the National Autonomous University of Mexico (UNAM)  
Participation in the *HAWC Collaboration Meeting* and the *Workshop on a wide field-of-view Southern Hemisphere TeV gamma ray observatory*

## Awards

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|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| 12/2020 | <b>Financial Performance Bonus for Active and Excellent Engagement in Research Discussions</b> - Deutsches Elektronen-Synchrotron (DESY) |
| 2013    | <b>DPG-Abiturpreis Physics</b> - High school award of German Physical Society                                                            |

## Teaching Experience

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|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| 07/2023 - 09/2023 | <b>DESY Summer student program</b>                                                                                       |
| 07/2022 - 09/2022 | Supervision of 3 summer student projects on GRB afterglow modelling                                                      |
| 10/2020 - 02/2021 | <b>Mathematical Tools for a Physicist at Humboldt University</b><br>Preparation and discussion of weekly exercise sheets |

## Outreach

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022, 2021  | <b>International Cosmic Day (yearly event)</b><br>Talk and supervision of a cosmic-ray experiment in German high school                                                                                                                                                    |
| 02/2022     | <b>Attended Workshop on Outreach Skills</b><br>3-day workshop of the German network of (Astro)particle physics outreach                                                                                                                                                    |
| 2018 - 2020 | <b>AIS<sup>3</sup> - Astroparticle Immersive Synthesizer<sup>3</sup></b><br>Construction and Maintenance of <a href="#">AIS<sup>3</sup></a> with Tim Otto Roth in Berlin (08-09/2018), Munich (02/2019) and Aachen (fall 2019) including outreach talks for school classes |

## Skills

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|------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Computer Skills</b> | Python, C++<br>LaTeX, Microsoft Office (Excel, Word, PowerPoint), Markdown, GIT |                                                       |
| <b>Languages</b>       | German ( <i>mother tongue</i> )<br>English ( <i>fluent</i> )                    | Spanish ( <i>advanced</i> )<br>Dutch ( <i>basic</i> ) |

## Intercultural Experience

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/2019           | <b>Attended Workshop on intercultural communication (2 days)</b><br>Combining theoretical concepts with practical application to train the ability to detect and resolve intercultural communicational problems |
| 11/2018           | <b>German Italian Physics Exchange (GIPE) to Trieste, Italy (4 days)</b>                                                                                                                                        |
| 10/2018 - 03/2019 | <b>Mentoring of international master students in Aachen</b>                                                                                                                                                     |
| 04/2016 - 09/2016 | <b>Support of a Spanish exchange student in Aachen (BeBuddy)</b>                                                                                                                                                |
| 11/2014           | <b>Intercultural training (2 days)</b>                                                                                                                                                                          |
| 06/2012 - 08/2012 | <b>School exchange to Instituto de Ballester, Buenos Aires, Argentina</b>                                                                                                                                       |

## Talks

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|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16/10/2024               | <b>NOVA Network 3 meeting, Nikhef Amsterdam</b><br>Lepto-hadronic modelling of gamma-ray burst afterglows                                                                                                      |
| 14/10/2024               | <b>GRAPPA journal club, GRAPPA Amsterdam</b><br>The Astrophysical Multi-Messenger Modelling (AM <sup>3</sup> ) Software                                                                                        |
| 21/02/2024               | <b>Invited: Workshop on Numerical Multi-messenger Modeling, Paris</b><br>News from AM <sup>3</sup>                                                                                                             |
| 28/11/2023               | <b>Invited: SPIMAX Seminar, Oxford</b><br>On the camel nature of GRB afterglows                                                                                                                                |
| 24/10/2023               | <b>Plenary: High Energy Phenomena in Relativistic Outflows (HEPRO)</b><br>Lepto-Hadronic Very-High-Energy GRB Afterglows                                                                                       |
| 30/06/2023               | <b>Gamma Group Seminar, DESY Zeuthen</b><br>GRB 221009A: Afterglow spectrum                                                                                                                                    |
| 21/02/2023               | <b>Invited: DESY Astroparticle division retreat, Potsdam, Germany</b><br>Intro-slides for critical review discussion on transients (with R. Konno)                                                             |
| 26/01/2023               | <b>Science Club of Multimessenger School (online, incl. Weizmann Institute, Israel)</b><br>The Weizsäcker-Williams Approach                                                                                    |
| 10/01/2023               | <b>Invited: Transient Tuesday, Niels Bohr Institute, Copenhagen (online)</b><br>VHE GRB Afterglows: A story about Bactrians, Dromedaries and lots of Butterflies                                               |
| 18/11/2022               | <b>Invited: Astroparticle Seminar at DESY, Zeuthen</b><br>VHE GRB Afterglows: A story about Bactrians, Dromedaries and lots of Butterflies                                                                     |
| 16/05/2022<br>18/05/2022 | <b>Talks at the Harnack-Haus GRB Workshop Berlin</b><br><ul style="list-style-type: none"> <li>• One Zone Basics and Effective Descriptions</li> <li>• Multi-wavelength Fitting at the Counts Level</li> </ul> |
| 27/10/2021               | <b>TeVPA 2021 (online due to Covid)</b><br>Understanding the Spectrum of Gamma-Ray Burst 190114C                                                                                                               |
| 23/09/2021               | <b>DESY Theory Workshop: Bright ideas for a dark universe (online due to Covid)</b><br>Understanding the Spectrum of Gamma-Ray Burst 190114C                                                                   |
| 22/04/2021               | <b>Science Club of Multimessenger School (online, incl. Weizmann Institute, Israel)</b><br>Relativistic Shocks                                                                                                 |
| 22/01/2020               | <b>Theory Group Seminar, DESY Zeuthen</b><br>Blazar Variability with Magnetic Reconnection                                                                                                                     |
| 26/03/2019               | <b>German Physical Society Meeting</b><br>FACT - Spectral Variability of TeV-Blazars                                                                                                                           |

## Posters

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|------------|----------------------------------------------------------------------------------------------------------------------------|
| 06/06/2023 | <b>Annual Meeting of the Multimessenger School in Israel</b><br>Understanding the Spectrum of Gamma-Ray Burst 190114C      |
| 12/05/2022 | <b>7th Meeting of the Astroparticle Physics Committee at DESY</b><br>Understanding the Spectrum of Gamma-Ray Burst 190114C |

## List of Publications

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>Leptohadronic Scenarios for TeV Extensions of Gamma-Ray Burst Afterglow Spectra</b><br>M. Klinger, C Yuan, A. M. Taylor and W. Winter ( <a href="#">ApJ 977 242</a> )                                |
| 2024 | <b>AM<sup>3</sup>: An open-source tool for time-dependent lepto-hadronic modelling of astrophysical sources</b><br>M. Klinger, A. Rudolph, X. Rodrigues, et al. (software, <a href="#">ApJS 275 4</a> ) |
| 2024 | <b>The Multiwavelength Picture of GRB 221009A's Afterglow</b><br>M. Klinger, A. M. Taylor, T. Parsotan, et al. ( <a href="#">MNRAS 529</a> )                                                            |
| 2023 | <b>Probing the multiwavelength emission scenario of GRB 190114C</b><br>M. Klinger, D. Tak, A. M. Taylor and S. J. Zhu ( <a href="#">MNRAS 520 1</a> )                                                   |
| 2020 | <b>FACT - Probing the Periodicity of Mrk 421 and Mrk 501</b><br>M. Klinger, M. Beck and F. Theißen on behalf of the FACT collaboration <a href="#">PoS(ICRC2019)</a>                                    |
| 2017 | <b>Observation of radio galaxies with HAWC</b><br>D. Avila, R. Alfaro, A. Galván, M.M. González, N. Fraija and M. Klinger on behalf of the HAWC Collaboration <a href="#">PoS(ICRC2017)</a>             |